

epository.

1. In the branch dropdown in the top left of the window, pick branch-A.

1. Open HelloWorld.py and click Edit.

1. Fix a code quality problem by replacing line 1 with this code:

```
python
def helloworld(a):
```

1. Fix another code quality problem by deleting line 3.

1. Commit these changes with the commit message Fix code quality problems

Task F. Compare the code quality of branch-A to the code quality of main

1. In the left navigation pane, click Build > Pipelines.

1. Wait for the most recent pipeline to show passed status. This might take as long as 5 minutes.

1. In the left navigation pane, click Merge requests. Click the Draft: Branch A MR to see the MR details page.

1. Half-way down the MR details page, if it says No changes to code quality, you may have to refresh the page. The pane should say Code Quality scans found 2 fixed findings.. This means you have fixed 2 code quality problems on branch-A which remain unfixed on main.

1. Expand the code quality pane to see the code quality problems you fixed on branch-A.

1. To transport or save the results of code quality scanning, the json artifact is available on the Pipelines page by clicking on the vertical ellipses right of the page.

Lab Guide Complete

You have completed this lab exercise. You can view the other lab guides for this course.

Suggestions?

If you wish to make a change to the Hands-On Guide for GitLab CI/CD, please submit your changes via Merge Request!

SECTION: customer-success/professional-services-engineering/education-services/gitlabpmhandson.md

GitLab Agile Portfolio Management Lab Guides

Lab Name	Lab Link
Access the GitLab training environment	Lab Link
Create an organizational structure in GitLab	Lab Link
Use GitLab planning tools	Lab Link
Create issues	Lab Link
Organize and manage issues	Lab Link
Use a merge request to review and merge code	Lab Link
Create and customize issue boards	Lab Link
Create and manage a Kanban board	Lab Link
Create and manage a Scrum board	Lab Link
Create and manage a Waterfall board	Lab Link

Quick links

Here are some quick links that may be useful when reviewing this Hands On Guide.

[GitLab Agile Portfolio Management course description](#)
[GitLab Agile Portfolio Management Specialist certification details](#)

Suggestions?

If you'd like to suggest changes to the GitLab Agile Portfolio Management Hands-on Guide, please submit them via merge request.

SECTION: customer-success/professional-services-engineering/education-services/gitlabpmlhandsonlab1.md

> Estimated time to complete: 5 to 10 minutes

Objectives

In this lab, you will access the training lab environment. This environment will allow you to try all of the GitLab Ultimate features, providing you with an environment to follow along with all of the lab steps.

Task A. Access your Training Group and Lab Environment

1. Navigate to <https://gitlabdemo.com/invite> in a web browser.

1. In the Invitation Code field, enter the invitation code provided by your instructor or in the LevelUp LMS.

1. Select Provision Training Environment.

> The login details for accessing the environment are slightly different between self-paced training and instructor-led training. Please follow the instructions that suit your needs.

Self-Paced Training

1. For self-paced courses, you will need your GitLab.com username. To find your username, navigate to Gitlab.

1. In the left sidebar, select your profile image.

1. In the resulting dropdown, you will see a value starting with @. This is your GitLab username.

1. After inputting your invitation code, the portal prompts you for your GitLab.com username. Enter your GitLab.com user (excluding the leading @ sign) in the field provided.

1. Select Provision Training Environment.

1. Select My Group at the bottom of the page.

1. Sign in with your GitLab.com credentials.

1. You will be redirected to a My Test Group group that provides a sandbox for you to perform training lab steps in.

> This group has a GitLab Ultimate license to see all of the features while your personal username namespace requires a paid subscription or a free trial to access all of the features.

>

> If you receive a 404 error when accessing your group, it is likely due to the username input during the lab provisioning. Double check your GitLab username to make sure it is entered correctly.

1. From your My Test Group training subgroup, click the New project button.

1. Continue to Task B.

Suggestions?

If you'd like to suggest changes to this lab, please submit them via merge request.

SECTION: customer-success/professional-services-engineering/education-services/gitlabpmhandsonlab10.md

> Estimated time to complete: 30 minutes

Objectives

The Kanban and Scrum boards in previous labs showed the progress of issues as developers worked on them. A waterfall board shows the health of all the issues that your team is responsible for within in a single waterfall stage. With a pure waterfall workflow, all issues are due at the same time--at the end of the waterfall stage--so what you're tracking with the board is how

likely they are to be done by that date.

Task A. Make a milestone representing a waterfall stage

1. Navigate to the Awesome Software project.
1. In the left pane, click Plan > Milestones.
1. Click New milestone.
1. Type Release 2.0-Dev in the Title field.
1. Select today's date as the Start Date by clicking on today's date in the calendar.
1. Select 1 month from today's date as the Due Date by clicking on today's date in the calendar.
1. Write up a clear description for the milestone.
1. Click Create milestone.

Task B. Add issues to the milestone

1. In the left pane, click Issues.
1. Click Bulk Edit.
1. Select a few issues of your choice by clicking the faint checkbox to the left of the search bar above the list of issues.
1. In the right pane, select Release 2.0-Dev from the Milestone dropdown.
1. At the top of the right pane, click Update all to apply the label to all selected issues.

Task C. Simulate past work in the waterfall stage by setting issue health statuses

1. For each issue in the milestone, choose one of the scoped labels to apply: Health::On Track, Health::Needs Attention, or Health::At Risk.
- > Remember, to apply a label to an issue, click on the issue. Then, on the right side of the screen next to the Labels keyword, click on Edit. From there, apply one of the labels by clicking on the label name, and then clicking off to set it.

Task D. Make a board to track the health of issues in a single waterfall stage

1. In the left pane, click Plan > Issue Boards.
1. Using the dropdown at the top of the page, create a new board.
1. Title the board Release 2.0 Dev Stage

1. Deselect the Show the open list checkbox.

1. Leave the Show the Closed list checkbox selected.

1. Scope the board to display issues in the Release 2.0-Dev milestone.

1. Click Create board.

> We will now create three new lists for the board: a list with a Health::On Track label, a list with a Health::Needs Attention label, and a list with a Health::At Risk label.

1. In the top right corner, click Create list.

1. Check that the Label radio button is selected in the Scope section of the list configuration pane.

1. Open the Value drop-down, and select Health::On Track.

1. Click Add to board.

1. In the top right corner, click Create list.

1. Check that the Label radio button is selected in the Scope section of the list configuration pane.

1. Open the Value drop-down, and select Health::Needs Attention.

1. Click Add to board.

1. In the top right corner, click Create list.

1. Check that the Label radio button is selected in the Scope section of the list configuration pane.

1. Open the Value drop-down, and select Health::At Risk.

1. Click Add to board.

> At this point, you have a complete board set up, but it is still recommended that you practice using this board as you would in production. For example, practice dragging and dropping your issues across different lists as their at-risk status increases or decreases. Or, simulate completing some issues by dragging them into the Closed list. Make sure to click on the issues themselves to confirm they are closed.

Suggestions?

If you'd like to suggest changes, please submit them via merge request.

SECTION: customer-success/professional-services-engineering/education-services/gitlabpmhandsonlab2.md

> Estimated time to complete: 30 minutes

Objectives

In GitLab, projects and subgroups help you organize your codebases and effectively manage your projects. In this lab, you will learn how to create an organizational subgroup, how to create a project, and how to add a user to a project.

Organization Namespaces: You can create subgroups under any other groups that you have access to and have permissions to create groups underneath. Your organization should have best practices documented on how groups are structured or named. If you're not sure, you can locate another group's members and ask one of the users with Owner role who should be able to help.

Personal Namespaces: Although you can create projects in your personal namespace (ex. <https://gitlab.com/exampleuser/example-project>), you cannot create groups in your personal namespace.

Security Warning: Any users that have access to parent or grandparent groups will have inherited permissions to this group. In the training environment, each student only has direct member access to their subgroup and does not have inherited member access to the parent groups. See the subgroup membership docs to learn more.

> Tip: Remember that the breadcrumbs are a good way to navigate between subgroups when you're creating a multi-level collection of groups and subgroups.

Prerequisites

1. You have already redeemed the invitation/voucher code.
1. You have signed into <https://gitlab.com> with your existing username.
1. You have clicked the My Group button and opened a new browser tab with the URL <https://gitlab.com/gitlab-learn-labs/environments/session-{id}/iu{iu}>. You should see My Test Group - iu{iu} as the title of the group.

Task A. Review groups and projects for GitLab's own source code

1. In a new browser tab, navigate to <https://gitlab.com/gitlab-org>. This group contains all of GitLab's product source code that is mostly open source and transparent based on our company values.
2. In the list of groups and project, click the linked name for the Frontend subgroup. You might have to scroll down or use pagination to find it on the second page.
3. Within the Frontend subgroup, you can see all of the projects and subgroups that exist as part of this group. Spend a few moments exploring some of these subgroups and projects.

Task B. Create organizational subgroups

1. In the lab environment tab, ensure that you are in your My Test Group group. If you are not in the My Test Group group (as in, you are on the <https://gitlab.com/gitlab-learn-labs> splash page) go to <https://gitlab.com/gitlab-learn-labs/environments/session-{id}/iu{iu}>, replacing the {id} with the invitation code you used, and the iu{iu} with your username.

1. Click the New Subgroup button in the top right corner.

1. In the Subgroup name field, type **Awesome Inc.**

1. The Subgroup URL shows which namespace hierarchy that this group will be created in.

- Do not change the Subgroup URL in this class. You will always create projects under <https://gitlab.com/gitlab-learn-labs/>... since this is your top-level organization group for these steps.

1. The Subgroup slug will automatically populate based on the name. Rename awesome-inc to awesome.

- Best Practice: This slug is what will appear as the project path, which means that it is both the URL path and the folder name when you run `git clone`. It is a best practice to shorten this when appropriate.

1. Ensure that the Visibility level is set to Private.

- > A private subgroup will ensure that the group and its projects can only be viewed by members. For more details of visibility levels, see the documentation.

1. Click the Create subgroup button.

- You should now see the group with the title Awesome Inc and your browser URL will reflect the path at <https://gitlab.com/gitlab-learn-labs/environments/session-{id}/iu/{awesome}>.

1. Congratulations! You have created your first group and are ready to create additional groups on your own. Within the Awesome Inc group, create subgroups for each of these teams at your awesome company.

- Software
 - Core (this is a subgroup of Software, not Awesome Inc)
 - Android
 - iOS
- Infrastructure
- Security

Task C. Create a new project

1. Navigate to the Awesome Inc > Software > Core subgroup that you just created.

1. Click the Create new project tile.

1. Click the Create blank project tile.

1. In the Project name field, enter Family Budget Calculator.

1. Note that the project URL shows /awesome/software/core in its path. This shows that the project is in the core subgroup.

1. Leave the project slug as the default value, family-budget-calculator.

1. Set the Visibility Level to private.

- > See the visibility docs to learn more. Keep in mind that any parent groups that have Private visibility will be inherited for all child groups and projects and cannot be changed.

- >

- > Training Environment Limitation: In the training environment, your group is part of a private parent group so you will not be able to make a group or project Public. To test working with public projects, you can create a project in your personal namespace.

1. Enable the Initialize repository with a README checkbox.

1. Leave Enable Static Application Security Testing (SAST) unchecked.

1. Select Create project.

Task D. Add a project member and set their role

1. In the Family Budget Calculator project, click Manage > Members in the left sidebar.

1. Click the Invite members button in the top right corner.

1. Search for and select your instructor as the user you are inviting. If you are taking the self-paced course, enter your own email address.

1. In the Select a role dropdown, select Developer. If you are taking the self-paced course, select Owner.

- > See the permissions and roles docs to learn more about each role level.

1. Click the Invite button.

1. Refresh the page to see the user invited as a Direct Member.

- > Users will inherit permissions from the parent group hierarchy that this project exists in. If a user already has a higher level of access in a parent group (ex. Maintainer), that access level supersedes a lower level of permission assigned at the project level (ex. Developer).

Suggestions?

If you'd like to suggest changes, please submit them using a merge request.

SECTION: customer-success/professional-services-engineering/education-services/gitlabpmhandsonlab3.md

> Estimated time to complete: 45-60 minutes

Objectives

To help you organize your work, GitLab provides epics, iterations, and milestones. In this lab, you will learn how to create, configure, and manage epics, iterations, and milestones in your projects.

Task A. Review planning features in GitLab

1. In a new browser tab, navigate to the GitLab project source code at <https://gitlab.com/gitlab-org/gitlab>. Note the namespace structure indicated by the top of the page. You should be in the GitLab project inside the GitLab.org group.
1. At the top of the page above the project name, click GitLab.org to navigate to the parent group.
1. Note the number of epics, issues, and merge requests indicated in the left pane. These numbers represent work items across all subgroups and projects inside Gitlab.org.
1. In the left pane, click Epics. This takes you to a searchable list of epics in GitLab.org and all its subgroups.
1. In the left pane, click Plan > Roadmap. Epics and milestones in a group containing a start date or due date can be visualized in a form of a timeline (that is, a Gantt chart). The Roadmap page shows the epics and milestones in a group, one of its subgroups, or a project in one of the groups.
1. On the epic bars, you can see each epic's title, progress, and completed weight percentage. When you hover over an epic bar, a popover appears with the epic's title, start date, due date, and weight completed.
1. You can expand epics that contain child epics to show their child epics in the roadmap. You can select the chevron (v) next to the epic title to expand and collapse the child epics.

Task B. Create an epic

> Epics provide a way to organize and manage a set of issues and sub-epics that share a strategic theme. In addition to logical grouping, epics enable project managers to perform higher level planning and build a roadmap with visual status tracking. You can read more about epics in the documentation.

1. In the lab environment browser tab, at the top of the page, click the ... beside GitLab Learn Labs.

1. Click on My Test Group.

1. On the groups page, click on the Awesome Inc subgroup. Inside the subgroup, click Software.

1. In the left pane of the Software group landing page, click Epics.

1. In the upper-right corner above the search bar, click on the New epic button.

1. On the New Epic page, enter Feature Category: Retirement Planning in the Title (required) field. Having a descriptive title helps make it clear what issues and sub-epics should be associated with the epic.

1. In the Description field, paste the following:

markdown
Overview

This is the top-level epic for all features in the Retirement Planning category of Awesome Co.'s personal finance software.

Useful Links

- To-Do: add link to the feature strategy document
- To-Do: add contributing team member information

> There is no need to change any other options for the epic, but it is still important to understand the options.

1. There are additional options that you can configure or leave at their default setting.

- Confidentiality: Create private epics for internal use if you have a public facing project.

- Label: Apply labels to your epic, which are metadata tags that can be used to sort and filter your epics.

- Start Date and Due Date: Schedule your epics around project milestones.

- Color: Customize the display color of the epic when used in roadmap timeline bars. Look at the GitLab.org group's Roadmap in Task A for an example.

Task C: Create a child epic

> A parent epic can have multiple child epics. Use child epics to organize more complex topics into smaller, more focused epics and issues.

1. You will now create a new child epic to link to the parent epic. Navigate to your Epics page by clicking on Plan > Epics on the left pane.

1. Click the New epic button in the top right corner.

1. In the title section, type in Investment Tracking.

1. In the Description field, paste the following:

```
markdown
Overview
```

This epic tracks all work on Investment Tracking features and integrations, as part of the overall Retirement Planning category strategy.

1. Leave all other fields as they are, and click Create epic.

1. Return to the Software group's full list of epics by clicking Epics in the breadcrumbs at the top of the page.

1. Click into the Feature Category: Retirement Planning epic.

1. You will now designate the Investment Tracking epic as a child of the Retirement Planning epic. In the Child issues and epics tab, select the Add drop-down menu, and click Add an existing epic.

> You can also use this menu to create a child epic from scratch.

1. Type & in the field provided, and select Investment Tracking from the list of epics.

> The & symbol lets you refer to epics by their unique epic ID rather than their name. This is useful for when you have a large number of epics with similar names.

1. Click Add to link Investment Tracking as a child epic to the Retirement Planning parent epic.

Task D. Set milestones to represent product goals

> Milestones in GitLab are a way to track issues and merge requests created to achieve a broader goal in a certain period of time. Milestones allow you to organize epics, issues and merge requests into a cohesive group, with an optional start date and an optional due date. You can read more about milestones in the documentation.

1. Navigate to the Awesome Inc subgroup.

1. In the left pane, click Plan > Milestones. This will take you to the Milestones page.

1. Click New milestone in the top right corner.

1. In the Title section, type in Organization Kickoff.

1. In the Start Date section, use the calendar to enter in today's date.

1. In the End Date section, use the calendar to enter 2 days from today's date as the milestone end date.

> While a description is not necessary, it is recommended to type in a description to help clarify what the milestone is intended to be used for.

1. Click Create milestone.

1. In the breadcrumbs at the top of the page, click Milestones.

1. Click New milestone to create a second milestone.

1. In the Title section, type in Backend services deployed.

1. In the Start Date section, use the calendar to enter in today's date.

1. In the End Date section, use the calendar to enter 2 weeks from today as the milestone end date.

1. Click Create milestone.

1. In the breadcrumbs at the top of the page, click Milestones to view your newly created milestones.

You will later assign tasks to the epics and milestones you created, allowing you to use roadmaps to view the progress of your initiatives.

Task E. Schedule iterations as team sprints

> Iterations are mutually exclusive timeboxes intended to track team velocity, while milestones can represent larger, overlapping product goals. You can read more about iterations in the documentation.

1. In your subgroup structure, navigate to Awesome Inc > Software.

1. In the left pane, click Plan > Iterations.

1. Click New iteration cadence in the top right corner.

1. In the Title section, type in Team sprints .

1. In the Description section, type in Tracking team progress toward minimum viable product .

1. In the Automatic start date input, use the calendar to enter in today's date.

1. In the Duration section, select 2 weeks as the duration of each iteration.

1. In the Upcoming iterations section, select 6 for the number of upcoming iterations.

1. Ensure the Enable roll over checkbox is selected.

> At the end of the current iteration, all open issues are added to the next iteration when Enable roll over is selected.

1. Select Create cadence.

You will later assign individual tasks to iterations (sprints).

Task F. Create a wiki for project documentation

> If you don't want to keep your documentation in your repository, but you want to keep it in the same project as your code, you can use the wiki GitLab provides in each GitLab project. Every wiki is a separate Git repository, so you can create wiki pages in the web interface, or locally using Git. GitLab wikis support Markdown, Rdoc, AsciiDoc, and Org for content. Wiki pages written in Markdown support all Markdown features, and also provide some wiki-specific behavior for links. You can read more about wikis [here](#).

1. Navigate to your Family Budget Calculator project in the Software > Core subgroup.

1. In the left pane, click Plan > Wiki.

1. Click Create your first page button in the middle of the page.

1. In the Title section, type in Family Budget Calculator Documentation as the page title.

1. Paste the following text in the content field underneath the title field.

```
markdown
Summary
```

The Family Budget Calculator helps households stay on budget and save for the future.

Contact

Contact <YOUR-NAME> with questions or comments.

1. Click Create Page. You should see a new Wiki page with the content you created.

> If you'd like, edit the Family Budget Calculator Documentation page to add additional content or create additional wiki pages.

Suggestions?

If you'd like to suggest changes, please submit them via merge request.

SECTION: customer-success/professional-services-engineering/education-services/gitlabpmlab4.md

> Estimated time to complete: 45 minutes

Objectives

Issues are a core building block in GitLab that enable collaboration, discussions, planning and tracking of work. Issues are defined in the scope of a Project, not a Group. You can read more about them in the documentation.

Task A. Create and manage labels

> You can use labels to categorize epics, issues, and merge requests using colors and descriptive titles like bug, feature request, or docs. This allows you to dynamically filter and manage epics, issues, and merge requests. You can read more about them in the documentation.

1. Navigate to your Awesome Inc group.

1. In the left pane, click Manage > Labels.

1. Click New label in the upper right hand corner.

1. In the Title field, enter Status::Open. The 2 colons in the label title mean that this will be a scoped label.

> A scoped label uses a double-colon (::) syntax in its title, for example: workflow::in-review. An issue, merge request, or epic cannot have two scoped labels, of the form key::value, with the same key. If you add a new label with the same key but a different value, the previous key label is replaced with the new label.

1. In the Description field, type Item that is ready to begin work .

1. For the label's color, GitLab supports any hex color code. For this label, choose Blue-gray from the suggested color palette (or type 6699cc in the Background color field).

1. Click Create label.

1. Create the following additional labels, setting a description and background color of your choosing. Note that some of these are scoped and some are unscoped.

- Status::Open
- Status::WIP
- Status::Done
- Priority::High
- Priority::Medium
- Priority::Low
- Dev
- QA
- Security

1. Go to your Family Budget Calculator project. The project is inside the Awesome Inc > Software > Core group hierarchy.

1. Click Manage > Labels from the left sidebar.

1. Click the star icon to the left of the Subscribe button to designate the following labels as

prioritized labels. Prioritized labels appear at the top of your labels list.

- Priority::High
- Priority::Medium
- Priority::Low

Task B. Create issues for tracking work

1. In your Family Budget Calculator project, click Plan > Issues from the left sidebar.

1. Click the New issue button.

1. In the title section, type Third-party financial services integration.

1. The description is an optional section, but feel free to type in your own description for this issue.

1. Using the Assignees dropdown, assign the issue to yourself by clicking on the dropdown, and then clicking on your username. While we will leave the options as they are for now, it is important to understand what they do:

- Epic: Associates the issue with an epic.

- Milestone: Milestones in GitLab are a way to track issues and merge requests created to achieve a broader goal in a certain period of time.

- Labels: Apply labels to your issue, which are metadata tags that can be used to sort and filter your issues.

- Weight: Apply a weight value to your issue to measure the time, complexity, or value a given issue has or costs.

- Due date: Use in issues to keep track of deadlines and make sure features are shipped on time.

- Iteration: Associate the issue with an iteration to track it over a period of time. This allows teams to track velocity and volatility metrics.

1. Click the Create issue button.

1. In the issue metadata pane, click Edit next to the Labels field.

1. Select the Status::Open label, then click away from the metadata pane to apply the label to the issue.

1. Repeat the previous 2 steps to apply the Priority::Medium and Dev labels to the issue.

1. In the left pane, click Plan > Issues. You will see the issue you just created in the list along with its labels.

1. Create a second issue by clicking New issue in the top right of the issue list page.

1. In the Title section, type Backend services.

1. Paste the following in the Description section:

markdown

- Create DB
- Create service infrastructure
- Write documentation

1. Using the Assignees dropdown, assign the issue to yourself by clicking on the dropdown, and then clicking on your username.

1. Click the Create issue button.

1. Apply the following labels to the Backend services issue by clicking on the label, then click away from the metadata pane to apply the label to the issue: Dev, Status::Open, and Priority::High.

1. In the left pane, click Plan > Issues to see both issues with their labels.

1. Create a third issue by clicking New issue in the top right of the issue list page.

1. In the Title section, type Frontend services.

1. Paste the following in the Description section:

markdown

- UX design
- Integration
- Write documentation

1. Using the Assignees dropdown, assign the issue to yourself by clicking on the dropdown, and then clicking on your username.

1. Click Create issue.

1. Apply the following labels to the Frontend services by clicking on the label, then click away from the metadata pane to apply the label to the issue: Dev, Status::WIP, and Priority::High.

1. In the left pane, click Plan > Issues to see all 3 issues with their labels.

Suggestions?

If you'd like to suggest changes, please submit them via merge request.

SECTION: customer-success/professional-services-engineering/education-services/gitlabpnhandsonglab5.md

> Estimated time to complete: 45-60 minutes

Objectives

To help you manage your issues, GitLab provides metadata for each issue. Metadata allows you to define details for an issue such as weight, and due dates. In this lab, you will learn how to view and manage issue metadata.

- Epic: Associates the issue with an epic.
- Milestone: Milestones in GitLab are a way to track issues and merge requests created to achieve a broader goal in a certain period of time.
- Labels: Apply labels to your issue, which are metadata tags that can be used to sort and filter your issues.
- Weight: Apply a weight value to your issue to measure the time, complexity, or value a given issue has or costs.
- Due date: Use in issues to keep track of deadlines and make sure features are shipped on time.
- Iteration: Associate the issue with an iteration to track it over a period of time. This allows teams to track velocity and volatility metrics.

You can learn more about issues and metadata in the documentation.

Task A. Set issue metadata

1. In your Family Budget Calculator project, click Plan > Issues from the left pane.
1. Click on the Third-party financial services integration issue that you created in a previous lab.
1. In the issue's metadata pane (on the right side of the screen), click Edit next to the Epic field.
1. Assign this issue to the Investment Tracking epic by clicking on the Investment Tracking in the list.
1. In the issue's metadata pane (on the right side of the screen), click Edit next to the Iteration field.
1. Assign this issue to the first most recent iteration in Team sprints by clicking on it in the list.

1. In the issue's metadata pane (on the right side of the screen), click Edit next to the Due date field.

1. Set the issue due date to 1 week from today's date by using the calendar.

1. In the issue's metadata pane (on the right side of the screen), click Edit next to the Labels field.

1. Apply the label Status::WIP. Note this replaces the previous Status::Open label, since an issue can't simultaneously have multiple labels with the same scope (the "Status::" part of the label).

> For more information about scoped labels, see the documentation.

1. In the issue's metadata pane, click Edit next to the Weight field.

1. Enter a value of 2, then click away from the metadata pane to set the issue's weight.

1. Navigate to Awesome Inc > Software group. In the left pane, click Epics. Click the Investment Tracking epic. Note that the Third-party financial services integration issue appears in the epic's details page.

1. In the left pane, click Plan > Iterations. Click the iteration to which you assigned the issue. Note that the Third-party financial services integration issue appears on the iteration's details page.

Task B. Organize, promote, and link issues and epics

1. In your Family Budget Calculator project, click Issues in the left pane.

1. Click your Backend services issue. You realize that, considering the scope of this initiative, this issue would be better suited as an epic, with each To-Do in the description as a separate issue in the epic.

1. To promote this issue to an epic, use the /promote quick action in the issue's comment field, then click Comment. If the new look for issues is enabled, use /promoteto epic quick action instead.

> A quick action is a text-based shortcut for common actions that are usually done by selecting buttons or dropdowns in the GitLab user interface. You can enter these commands in the descriptions or comments of issues, epics, merge requests, and commits. For more information about quick actions, click [here](#).

>

> You can also promote an issue to an epic by clicking the vertical ellipsis next to the Edit button, then clicking Promote to epic in the resulting menu.

1. After applying the quick action, Backend services is now an epic at the Core group level. Note that there is no comment that says '/promote', as quick actions do not leave a comment. Using the breadcrumbs at the top of the page, click your Core subgroup.

1. In the left pane, click Epics.

1. Click into your new Backend services epic.

1. You'd like to link the epic's To-Do items as individual issues. While you can promote issues into epics, you cannot promote To-Do items into an issues. Instead, you will need to create a new issue for each To-Do item. Under the Child issues and epics tab, click Add > Add a new issue.

1. Enter Create DB as the issue title.

1. In the Project dropdown, select Family Budget Calculator, as it is the only project created so far, and all issues must belong to a project.

1. Click Create issue in the top right corner.

1. Follow the previous four steps to create the Create service infrastructure and Backend documentation issues, both linked to the Backend services epic.

1. Click on the Create DB issue.

1. In the issue's right metadata pane, assign the issue to the Backend Services Deployed milestone.

1. Assign the Create service infrastructure and Backend documentation issues to the Backend Services Deployed milestone. So far all issues have been created in the Family Budget Calculator project by necessity. But as requirements grow it may be best to move some issues into more suitable projects.

1. Navigate to your Software > Core subgroup.

1. From the Core group landing page, click New project, then Create blank project.

1. In the Project name field, enter Database

1. Leave the Visibility Level at the default selection.

1. Enable the Initialize repository with a README checkbox.

1. Click Create project.

1. Return to your Family Budget Calculator project in the Core subgroup.

1. In the left pane, click Issues.

1. Click into the Create DB issue.

1. On the issue's details page, scroll to the bottom of the right metadata pane and click Move issue.

1. Select your Database project, then click Move.

1. See that the project heading and breadcrumbs at the top of the page indicate your issue is now part of the Database project.

Task C. Create and apply a description template

1. Navigate to your Awesome Inc group by using the breadcrumbs at the top of the page.

1. In the upper right corner of the group landing page, click New Project.

1. Click Create blank project.

1. Enter Description Templates in the Project name section. This project will hold templates that can be used to pre-populate issues and merge requests across the organization.

1. Enable the Initialize repository with a README checkbox, as we are not importing in a project that already exists from elsewhere. Leave all other options as their default.

1. Click Create project.

1. On the project landing page, click the (+) dropdown near the top of the page under the project title.

1. Click This Directory > New file.

1. In the File name field, enter .gitlab/issuetemplates/technicalspike.md

1. For the file's content, paste the following:

```
markdown
Instructions
Use this issue to capture research that must take place before continued development of a feature.
```

```
Summary
<!--In 2 sentences or fewer, describe the problem to be solved or the question to be answered. -->
```

```
Impact Statement
<!-- Describe importance of solving the problem. How will it affect the feature or product direction? -->
```

Tasks

- [] Assign participants and DRI
- [] Apply appropriate priority and team labels
- [] Assign to an upcoming product sprint

```
/label ~"Status::Open"
```

> Notice that the template is using a quick action to assign a label. Whenever this template is used, a label will automatically be assigned when it is created.

1. Click Commit changes.

> Security Warning: Making direct commits on the main branch without a merge request is not recommended for enterprise level projects.

1. Using the breadcrumbs at the top of the page, navigate back to your Awesome Inc group.

1. In the left pane, click Settings > General.

1. Scroll to the Templates section and select Expand.

1. In the Templates dropdown, select your Description Templates project.

1. Click Save changes.

1. Now the template can be applied when creating an issue. Navigate to your Awesome Inc > Software > Core > Database project.

1. In the left pane, click Plan > Issues.

1. Click New issue.

1. In the Title field, type Identify tuning parameters to reduce performance bottlenecks

1. In the Description field, expand the Choose a template dropdown and select your technicalspike description template.

1. (Optional): Edit the description to fill in more details about the issue.

1. Assign the issue to yourself by clicking on the Assign to me link and click Create issue.

1. Review the pre-populated description and metadata on the issue's details page.

Suggestions?

If you'd like to suggest changes to the merge request, please submit them via merge request.

SECTION: customer-success/professional-services-engineering/education-services/gitlabpmhandsonlab6.md

> Estimated time to complete: 45-60 minutes

Objectives

A merge request is a proposal to incorporate changes from a source branch to a target branch. Merge requests help you manage the changes that are applied to your code. In this lab, you will learn how to setup and manage merge request approval rules in your projects. You can learn more about merge requests in the documentation.

Approval rules define how many approvals a merge request must receive before it can be merged, and which users should do the approving. They can be used in conjunction with Code owners to ensure that changes are reviewed both by the group maintaining the feature, and any groups responsible for specific areas of oversight. See the documentation to learn more.

Task A. Set merge request approval rules

1. Navigate to your Database project inside the Software > Core subgroup.

1. In the left pane, click Settings > Merge Requests.

1. Scroll down to the Merge request approvals section and click Add approval rule.

1. In the Rule name field, enter Infra team

1. In the Groups field, select your Infrastructure group. You may need to use the 'All groups' option and search for your top-level group.

> To help reduce the search results, try searching for /awesome/infrastructure, or yourgroupname/awesome/infrastructure.

1. Click Save changes.

1. Back in the Merge request approvals section of the Merge Requests page, click Add approval rule to create a second project-level rule.

1. In the Rule name field, enter Security operations

1. In the Groups field, select your Security group. You may need to use the 'all groups' option and search for your top-level group.

1. Click Save changes.

1. Back in the Merge request approvals section of the Merge Requests page, under Approval Settings, check the box next to Prevent editing approval rules in merge requests.

1. In the same section, ensure that Prevent approval by the author is unchecked. This is necessary so we can approve our own merge requests in the training environment.

1. Click Save changes.

Task B. Create a merge request

1. In your Database project, click Issues in the left pane.

1. Click into your Identify tuning parameters to reduce performance bottlenecks issue.

1. Click the down-arrow dropdown next to the Create merge request button on the issue landing page.

1. Ensure Create merge request and branch is checked.

1. In the Branch name field, change the text to read update-db-docs-perf-tools.

1. Ensure that the Source is set to main.

1. Click Create merge request.

1. Type Draft: Add performance tools to database documentation in the Title field.

> Putting 'Draft:' at the beginning of your merge request means that the merge request will not occur until it has been marked as ready. This is used to note that a merge request is not ready to be merged yet, and to prevent accidental merges. Note that the Mark as draft checkbox below the title will also check automatically when 'Draft:' is added to the title.

1. Remove Closes <ISSUE NUMBER> from the Description field. We want to keep the original issue open for additional work.

> If a merge request has 'Closes <ISSUE NUMBER>' in their description, the issue will be closed when the merge request is merged.

1. Verify that you are assigned to the merge request by checking the Assignees section. Also note any labels inherited from the issue, and any approval rules inherited from project settings.

1. Click the Create Merge Request button.

1. From the merge request details page, select Code > Open in Web IDE to edit files on the update-db-docs-perf-tools branch.

> The Web IDE is an advanced editor with built-in ability to commit to your repository branches. You can use the Web IDE to make changes to multiple files directly from the GitLab UI. See the documentation to learn more.

1. Click README.md from the left file pane.

1. Paste the following into README.md, beginning on line 3.

```
markdown
Performance tools
The database currently uses HAProxy for load balancing.
We are researching and testing additional tools to improve performance.
```

1. Click Source Control in the left pane (the third button from the top).

1. In the Commit message field, enter Update docs with performance tools

1. Ensure that the message "Commit and push to update-db-docs-perf-tools branch" is written in the red 'commit' box underneath your commit message.

1. Click Commit and push to update-db-docs-perf-tools branch.

Task C. Perform code review and merge changes

> For advice on best practices when it comes to code reviews, see the documentation.

1. Navigate to the Draft: Add performance tools to database documentation merge request by clicking on the red GitLab button in the bottom left corner of the Web IDE, and then clicking Go to Database project on GitLab.

1. Click on Code > Merge Requests, and then click on Draft: Add performance tools to database documentation.

1. On the merge request page, click the Changes tab to see the changes that will be applied to the project's main branch after merge.

> Code reviewers can critique individual lines of code and suggest changes. See the documentation to learn more.

1. Click on the Overview tab.

1. In the three dots menu to the right of the issue title, click Mark as ready to take the merge request out of draft mode. You will see that Draft: has been removed from the merge request's title.

1. Click Approve to approve the merge request. Note that the Merge button now appears since all requisite approvals have been applied.

1. Scroll down to the comments field and type a comment in the comments field that reads: Approved. Ready to merge.. Click Comment to post the comment.

1. Click Merge and observe the merge complete successfully.

1. Navigate to the project landing page by clicking the Database title tile in the top left corner. See that the README.md file on the main branch now includes your updates.

1. In the left pane, click Code > Merge requests. The merge request will now appear under the Merged tab on this page.

Suggestions?

If you'd like to suggest changes, please submit them via merge request.

SECTION: customer-success/professional-services-engineering/education-services/gitlabpmhandsonlab7.md

> Estimated time to complete: 30 minutes

Objectives

To help you plan, organize, and visualize a workflow, you can utilize an issue board.

The issue board is a software project management tool used to plan, organize, and visualize a workflow for a feature or product release. You can use it as a Kanban or a Scrum board. Issue boards can be configured to meet the needs of various project management frameworks.

You can learn more in the documentation.

In this lab, you will learn how to create a simple issue board.

Task A. View and customize project-level Issue boards

1. Navigate to your Family Budget Calculator project in the Software > Core subgroup.

2. In the left pane, click Plan > Issue boards.

> The default Development board contains only 2 lists by default: issues with the Open status and issues with the Closed status. For this lab, we are going to add more lists that use issue labels.

3. In the top right corner, click New list.

4. Ensure the Label radio button is selected in the Scope section of the list configuration pane. You can also create lists that are based on other metadata.

- Assignee: Who is directly responsible for an issue.

- Milestone: The release/due date of issues.

- Iteration: The velocity of issues.

5. Open the Value drop-down, and select Status::WIP.

6. Click Add to board. All issues tagged with Status::WIP should appear in the new list.

7. Add another custom list to the project board: in the top right corner, click New list.

8. The new list will be scoped by issue label. Verify that the Label radio button is selected in the Scope section of the list configuration pane.

9. Open the Value drop-down, and select Status::Done.

10. Click Add to board.

> Your Development board should now include the custom lists Status::WIP and Status::Done along with the default lists Open and Closed.

11. Note the Create service infrastructure issue in the Open list. Use your mouse to drag the Create service infrastructure issue into the Status::WIP list.

12. Click into the Create service infrastructure issue card by clicking directly on the issue's title. Note the Status::WIP label was automatically applied to that issue when you dragged it to a new list.

13. On the issue details page, click the Edit button next to Labels in the metadata pane.

14. Select the Status::Done label, and click away from the metadata pane to apply the label.

15. Return to Plan > Issue boards using the left pane.

> The Create service infrastructure issue should now appear in the Status::Done list.

Task B. Manage group-level issue boards

> Boards can also be viewed and managed at the group level.

1. Using the breadcrumbs at the top of the page, navigate to your top level Awesome Inc group.

1. In the left sidebar, navigate to Plan > Issue boards.

> Group-level boards will have a similar default board to project-level boards. This group-level issue board shows all issues across the group's subgroups and projects.

1. At the top of the page, click the options toggle directly to the right of the search bar. Toggle Epic swimlanes on. The board view should refresh and show a swimlane view of lists grouped by epic.

1. Scroll down to the bottom of the page and expand Issues with no epic assigned.

1. Click and drag the Identify tuning parameters to reduce performance bottlenecks issue into your Backend services epic.

1. Hover your mouse over the Backend services heading. Click the Go to epic link from the details box that appears.

1. Verify your Identify tuning parameters to reduce performance bottlenecks issue is part of the Backend services epic.

Task C. Create a new issue board

1. Navigate to Plan > Issue boards.

1. At the top of the page, click the Development dropdown to access the Switch board menu.

1. Click Create new board.

1. Title the board <YOUR NAME>

1. Deselect the checkboxes next to Show the Open list and Show the Closed list. This will remove the default lists from your custom board.

1. Click the Expand button next to Scope.

1. Click Edit next to Assignee and select yourself.

1. Click Create board.

1. In the top right corner, click New list.

1. Check that the Label radio button is selected in the Scope section of the list configuration pane.

1. Open the Value drop-down, and select Priority::High.

1. Click Add to board.

1. In the top right corner, click New list.

1. Check that the Label radio button is selected in the Scope section of the list configuration pane.

1. Open the Value drop-down, and select Status::WIP.

1. Click Add to board.

1. In the top right corner, click New list.

1. Click on the Milestone radio button in the Scope section of the list configuration pane.

1. Open the Value drop-down, and select Backend services deployed.

1. Click Add to board.

1. Refresh the browser tab that contains your new board.

Task D: Create a new issue for your board

1. In the Priority::High list, click the (+) icon to create a new high-priority issue.

1. Title the issue Update family budget app personas

1. Select Family Budget Calculator as the project for the issue to belong to.

1. Click Create issue.

1. An issue details pane should appear on the right side of the page. Assign the issue to yourself if it is not already assigned. Add an additional Status::Open label to the issue.

1. Click the X in the top right corner to close out of the issue details pane.

1. Click the diagonal arrows in the top right of the page to enter Focus mode. The rest of the GitLab navigation UI is now hidden, allowing you to focus on issues in the board.

1. Click the diagonal arrow again to leave Focus mode.

Suggestions?

If you'd like to suggest changes, please submit them via merge request.

SECTION: customer-success/professional-services-engineering/education-services/gitlabpmhandsonlab8.md

> Estimated time to complete: 30 minutes

Objectives

Kanban boards show you the progress of all tickets in your project, as they move from the Backlog, to being worked on, to being checked by QA, to being closed. A real-world Kanban board might involve many more statuses than these 4, but these are adequate for a bare-bones Kanban demonstration. To learn more about Kanban boards, [click here](#).

Task A. Create a new subgroup to work in

1. In the GitLab environment, navigate to the Awesome Inc group.

1. Click on the New subgroup button in the top right.

1. Type PM Workflows into the Subgroup name field.

1. Click Create subgroup to create the subgroup.

Task B. Create epics

1. From your new subgroup, in the left pane, click Epics. We will be creating three epics.

1. Click the New Epic button in the top right.

1. Type Frontend into the Title (required) field. We will be leaving all the other fields of the epic blank.

1. Click Create epic.

1. Click the New Epic button in the top right.

1. Type Backend into the Title (required) field. We will be leaving all the other fields of the epic blank.

1. Click Create epic.

1. Click the New Epic button in the top right.

1. Type QA into the Title (required) field. We will be leaving all the other fields of the epic blank.

1. Click Create epic.

Task C. Use a project template with issues

> GitLab has a variety of project templates you can use to help jump start your projects. The one we will be using is called Sample GitLab Project, which will come with predefined issues. You can see all of GitLab's project template [here](#).

1. Click the PM Workflows title tile in the left pane to return to the subgroup landing page.

1. At the top of the page, click New project.

1. Click the Create from template option in the top right section.

1. Scroll to the bottom of the built-in templates. Next to Sample GitLab Project, click on Preview to see the project in a new tab. It is always important to preview a project before using it as a template.

1. After you have explored the template, return to your previous tab, and click Use template.

1. Type in Awesome Software the Title section. Leave all other options as their default values.

1. Click the Create project button. GitLab will now import a project containing many pre-generated issues and merge requests. When the import process has finished, you will be taken to the new project's landing page.

Task D. Assign some issues to epics

1. In the left pane, click Plan > Issues.

1. Click an issue of your choice.

1. Assign the issue to the Backend epic using the right metadata pane on the issue's details page.

1. Assign another issue of your choice to the Frontend epic using the right metadata pane on the issue's details page.

1. Assign another issue of your choice to the QA epic using the right metadata pane on the issues's details page.

Task E. Create labels representing Kanban stages

1. Using the breadcrumbs at the top of the page, return to your PM Workflows subgroup.

1. In the left pane, click Manage > Labels.

1. Click the New Label button in the top right.

1. Type in Status::Backlog into the title field.

1. Feel free to pick a color for the label and type in a description. When you are satisfied, click Create label.

1. Click the New Label button in the top right.

1. Type in Status::WIP into the title field.

1. Feel free to pick a color for the label and type in a description. When you are satisfied, click Create label.

1. Click the New Label button in the top right.

1. Type in Status::QA into the title field.

1. Feel free to pick a color for the label and type in a description. When you are satisfied, click Create label.

> You are not making a label to mark an issue as "done." Instead, you will close each issue when it is done. This lets GitLab register the issue as complete in burndown/burnup charts and on roadmaps.

1. Click the New Label button in the top right.

1. Type in Health::On Track into the title field.

1. Feel free to pick a color for the label and type in a description. When you are satisfied, click Create label.

1. Click the New Label button in the top right.

1. Type in Health::Needs Attention into the title field.

1. Feel free to pick a color for the label and type in a description. When you are satisfied, click Create label.

1. Click the New Label button in the top right.

1. Type in Health::At Risk into the title field.

1. Feel free to pick a color for the label and type in a description. When you are satisfied,

click Create label.

Task F. Put all issues in the Kanban backlog

> None of the issues have been worked on yet, so you need apply the Status::Backlog label to all of them. Fortunately you can perform bulk edits on issues.

1. In the left pane, click Issues.

1. Click Bulk Edit.

1. Select all the issues by clicking the faint checkbox to the left of the search bar above the list of issues.

1. In the right pane, select Status::Backlog from the Labels dropdown.

1. At the top of the right pane, click Update all to apply the label to all selected issues.

Task G. Create the Kanban board

1. In the left pane, click Plan > Issue Boards.

1. At the top of the page, click the Development dropdown.

1. Select Create new board from the Switch board menu.

1. Type Kanban in the Title section.

1. Deselect the Show the Open list checkbox, as we will not need it for this board.

1. Keep the Show the Closed list checkbox selected.

1. Kanban boards generally show all your issues, so don't set a scope for the board.

1. Click Create board.

Task H. Add a list for each stage

1. At the top right of the page, click Create list.

1. In the New list options, select Label as the list scope and Status::Backlog as the value.

1. Click Add to board.

1. At the top right of the page, click Create list.

1. In the New list options, select Label as the list scope and Status::WIP as the value.

1. Click Add to board.

1. At the top right of the page, click Create list.
1. In the New list options, select Label as the list scope and Status::QA as the value.
1. Click Add to board.
1. Refresh the browser page to force the 3 new lists to appear in the same order that you created them.

Task I. Limit the allowed amount of work in progress

1. Click the gear icon at the top of the Status::WIP list to open list settings.
1. Next to Work in progress limit, click Edit button and set the value to 3
1. Click the X in the top right corner to exit out of the list settings.

Task J. Simulate work on issues

1. Practice dragging issues between different lists. For example, place more than 3 issues in the WIP list, and notice the results.
1. At the top of the page, click the Preferences icon > Epic swimlanes toggle on.
1. At the bottom of the page, expand Issues with no epic assigned to see the full list of issues in the Backlog list.
1. Practice dragging more issues, both to different lists and to different epics.

Suggestions?

If you'd like to suggest changes, please submit them via merge request.

SECTION: customer-success/professional-services-engineering/education-services/gitlabpmhandsonlab9.md

> Estimated time to complete: 30 minutes

Objectives

In this lab you'll configure a Scrum board and populate it with the same issues you used for the Kanban board. A Scrum board is a board that is used to keep track of the progress in a sprint. To learn more about Scrum, click [here](#).

Task A. Make Scrum-specific labels

> Compared to Kanban boards, Scrum boards normally require 2 extra labels to represent the project and sprint backlogs.

1. Navigate to the Awesome Software project.
1. In the left sidebar, click Manage > Labels
1. Click the New Label button in the top right.
1. Type in Status::Project Backlog into the title field.
1. Feel free to pick a color for the label and type in a description. When you are satisfied, click Create label.
1. Click the New Label button in the top right.
1. Type in Status::Sprint Backlog into the title field.
1. Feel free to pick a color for the label and type in a description. When you are satisfied, click Create label.

Task B. Put all issues in the Project Backlog

1. In the left pane, click Issues.
1. Using the bulk issue edit technique you learned in the last lab, apply the Status::Project Backlog label to all issues. Because this is a scoped label, it will remove any other Status labels that are already applied to any issues.

Task C. Make an iteration (sprint)

1. Iterations can only exist at the group or subgroup level. Using the breadcrumb trail, navigate to the PM Workflows subgroup.
1. In the left pane, click Plan > Iterations.
1. Click New iteration cadence.
1. Type Sprint 6 in the Title section.
1. Select today's date via the calendar in the Start Date section.
1. Select 2 as the number of weeks in the Duration section.
1. Select 2 as the number of iterations in the Upcoming Iterations section.
- > For the purposes of this lab, we will not worry about rollover, so there is no need to mark it.
1. Click Create cadence.

Task D. Add some issues to the next sprint

> This step simulates the "sprint planning" ceremony, where your team decides which issues it will work on in the upcoming sprint.

1. Navigate to the Awesome Software project.

1. In the left sidebar, click Issues.

1. Pick a handful of issues that you want to work on in the next sprint and assign them to the Sprint 6 first iteration. You can do this either with the bulk issue edit feature (selecting only the relevant issues), or by visiting each issue's details page.

1. Apply the Status::Sprint Backlog label to those same issues. You can do this either with the bulk issue edit feature (selecting only the relevant issues), or by visiting each issue's details page.

Task E. Make a Scrum board for the next sprint

1. In the left pane, click on Plan > Issue Boards and view your existing Kanban from the last lab.

1. Click Group by > Epic if the board is not already showing that view.

1. At the top of the page, click the dropdown to view the Switch boards menu.

1. Click Create new board.

1. Title the new board Sprint 6

1. Deselect the Show the Open list checkbox, as we do not need to have an Open list for this kind of board.

1. Leave the Show the Closed list checkbox selected.

1. Next to Scope, click Expand.

1. In the scope pane, click Edit near Iteration option. Select Current iteration.

1. Click Create board.

> We will now create three new lists for the board: a list with a Status::Sprint Backlog label, a list with a Status::WIP label, and a list with a Status::QA label.

1. In the top right corner, click Create list.

1. Check that the Label radio button is selected in the Scope section of the list configuration pane.

1. Open the Value drop-down, and select Status::Sprint Backlog.

1. Click Add to board.

1. In the top right corner, click Create list.

1. Check that the Label radio button is selected in the Scope section of the list configuration pane.

1. Open the Value drop-down, and select Health::Needs Attention.

1. Click Add to board.

1. In the top right corner, click Create list.

1. Check that the Label radio button is selected in the Scope section of the list configuration pane.

1. Open the Value drop-down, and select Status::QA.

1. Refresh the browser page to force the 3 new lists to appear in the same order that you created them.

> At this point, you have a complete board set up, but it is still recommended that you practice using this board as you would in production. For example, practice dragging and dropping your issues across different lists as their at-risk status increases or decreases. Or, simulate completing some issues by dragging them into the Closed list. Make sure to click on the issues themselves to confirm they are closed.

Task F. View your sprint progress

1. In the left pane, click Plan > Iterations.

1. Click into your Sprint 6 iteration, and then click on the first week in the list.

1. Note the progress of issues in the burndown and burnup charts.

Suggestions?

If you'd like to suggest changes, please submit them via merge request.

SECTION: customer-success/professional-services-engineering/education-services/lms.md

Thought Industries LMS Tech Stack Guide

The Tech Stack single source of truth is the Tech Stack YAML and contains more detail about this app.

```
{{% tech-stack "Thought Industries LMS" %}}
```

Implementation

The LMS implementation for Professional Services is organized into multiple phases.

System Diagram

The Thought Industries LMS Implementation is a SaaS app and is not integrated with other GitLab systems.

```
mermaid
graph TD
  GC[GitLab Customer] -->|user authentication| LMS[LMS]
```

Data Model

The Data Model is not available and LMS is a closed system.

Integrations

The Thought Industries LMS Implementation is a stand-alone SaaS app and is not integrated with other GitLab apps.

Key Reports / Dashboards

All Dashboards and Reports are a part of LMS itself. No separate Sisense reports are available or planned.

SECTION: customer-success/professional-services-engineering/education-services/secessentialshandson.md

Security Essentials Lab Guides

Lab Name	Lab Link
Configure SAST, Secret Detection, and DAST	Lab Link
Enable and Configure Dependency Scanning	Lab Link
Enable and Configure Container Scanning	Lab Link
Enable and Configure License Compliance	Lab Link
Enable and Configure Coverage-Guided Fuzz Testing	Lab Link
Enable and Scan Using a Scan Execution Policy	Lab Link

Quick Links

Here are some quick links that may be useful when reviewing this Hands-On Guide.

- GitLab Security Essentials Course Description
- GitLab Security Specialist Certification Details

Suggestions

If you'd like to suggest changes to the GitLab Security Essentials Hands-on Guide, please submit them via merge request.

SECTION: customer-success/professional-services-engineering/education-services/secessentialshandson1.md

> Estimated time to complete: 45 to 60 minutes

Objectives

In this lab you'll enable SAST, Secret Detection, and DAST scans for a GitLab project.

After the scans run in a CI/CD pipeline, you'll view the results of all 3 scans. You'll mark a vulnerability for future action, and you'll dismiss a different vulnerability.

Finally, you'll fix a vulnerability on a branch and introduce a new vulnerability on that same branch, so you can view the differences in vulnerabilities between default and non-default branches.

Please take time to understand any code that you are asked to copy and paste in any lab. Ask your instructor to explain any code that's not clear.

Task A. Access your Lab Environment

1. Navigate to <https://gitlabdemo>.